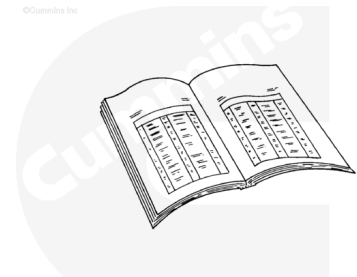


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



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- Disconnect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>)
- Remove the belt guard, if equipped. Refer to Procedure 008-001 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-001.html>)
- Remove the fan belt. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-002-tr.html>)
- Remove the fan and fan spacer(s). Refer to Procedure 008-040 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-040.html>)
- Remove the alternator belt and alternator. Refer to Procedure 013-001 in Section 13. (</qs3/pubsys2/xml/en/procedures/20/20-013-001-tr.html>)
- Remove the fan hub. Refer to Procedure 008-036 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-036-tr.html>)
- Remove the fan hub idler. Refer to Procedure 008-029 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-029-tr.html>)
- Remove the water pump. Refer to Procedure 008-062 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-062-tr.html>)

- Remove the water pump drive. Refer to Procedure 009-029 in Section 9.
(/qs3/pubsys2/xml/en/procedures/20/20-009-029-tr.html)
- Remove the thermostat housing. Refer to Procedure 008-015 in Section 8.
(/qs3/pubsys2/xml/en/procedures/20/20-008-015-tr.html)
- Remove the vibration damper. Refer to Procedure 001-052 in Section 1. (/qs3/pubsys2/xml/en/procedures/20/20-001-052-tr.html)
- Remove the front crankshaft seal. Refer to Procedure 001-023 in Section 1.
(/qs3/pubsys2/xml/en/procedures/20/20-001-023-tr.html)
- Remove the fuel pump. Refer to Procedure 005-016 in Section 5. (/qs3/pubsys2/xml/en/procedures/20/20-005-016-tr.html)
- Remove the front support bracket. Refer to Refer to Procedure 016-002 in Section
(/qs3/pubsys2/xml/en/procedures/20/20-016-002-tr.html) 16 (rail applications).
- Remove the oil scavenge pump. Refer to Procedure 007-060 in Section (/qs3/pubsys2/xml/en/procedures/20/20-007-060.html) 7 (rail applications).
- Remove the accessory drive pulley and assembly. Refer to Procedure 009-001 in Section 9.
(/qs3/pubsys2/xml/en/procedures/20/20-009-001-tr.html)

Rail Applications

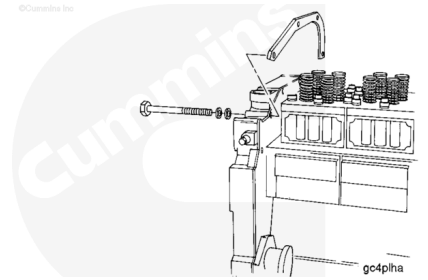
- Remove the front support bracket. Refer to Procedure 016-002 in Section
(/qs3/pubsys2/xml/en/procedures/20/20-016-002-tr.html) 16 (rail applications).
- Remove the oil scavenge pump. Refer to Procedure 007-060 in Section (/qs3/pubsys2/xml/en/procedures/20/20-007-060.html) 7 (rail applications).

Remove

Remove the gear cover clamping plate.

Note : Some engines have two plates.

This component is installed at the fan hub installation.



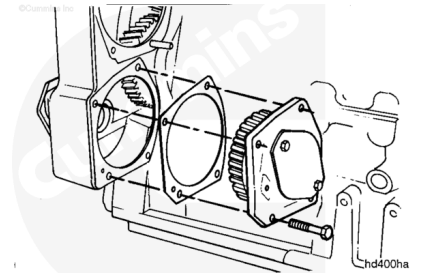
Remove the hydraulic pump drive.

Note : Some engines contain a cover plate instead of a hydraulic pump drive assembly.

Remove the four capscrews.

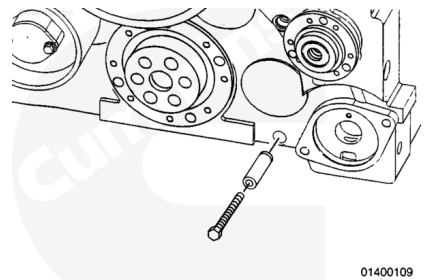
Remove the hydraulic pump drive assembly.

Remove and discard the gasket.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



The master dowel **must** be removed before the gear cover can be removed.

Use a 5/16-18 capscrew or a threaded rod. Use an old piston pin or a slide hammer. Remove the dowel.

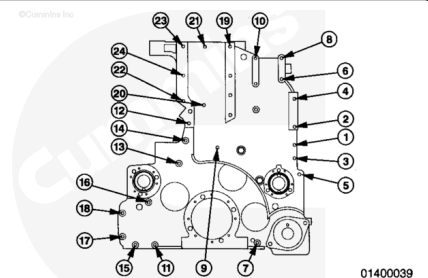
The front cover contains five capscrews on the bottom.

Remove the capscrews from the oil pan adapter.

The gear cover has 21 capscrews in the front of the cover and three, numbers 1, 2, and 3, on the back side of the cover.

Remove the 24 capscrews, four nuts, washers, and the gear cover.

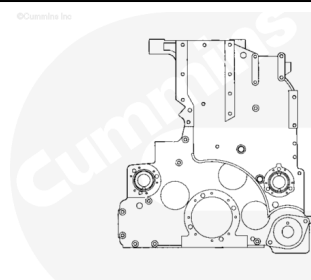
Remove and discard the gasket.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent or steam. Clean the front cover and related parts.

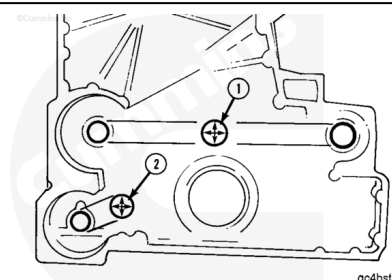
Measure the inside diameter of the bushings. The bushings **must** be replaced if **not** within specifications.

Accessory Drive & Water Pump Drive Bushing Inside Diameter

mm		in
39.75	MIN	1.565
39.90	MAX	1.571

Hydraulic Pump Drive (2) Bushing Inside Diameter

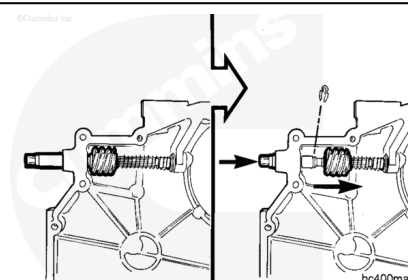
mm		in
38.13	MIN	1.501
38.25	MAX	1.506



Remove and check the barring mechanism.

Push the shaft in. Hold in position. Slide the worm gear until the spring is compressed.

Use a screwdriver. Remove the retaining ring.



Remove the parts:

- Shaft

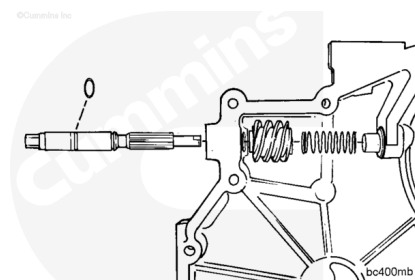
- Spring
- Worm gear.

Remove the o-ring from the shaft. Discard the o-ring.

Use solvent to clean the parts. Flush the oil drillings. Dry with compressed air.

Note : Some contact marks on gear are acceptable.

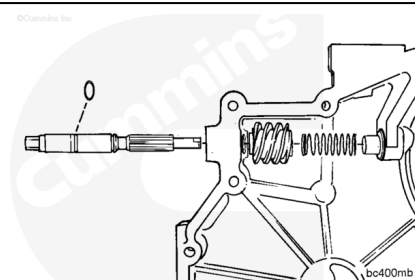
Check the parts for severe damage.



Use vegetable oil to lubricate the o-ring. Install the o-ring on the barring shaft.

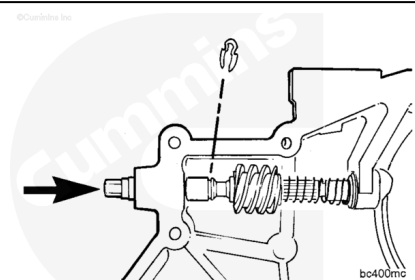
Install the parts:

- Worm gear
- Spring
- Shaft.



Install the retaining ring.

Rotate the shaft. Check for correct assembly.

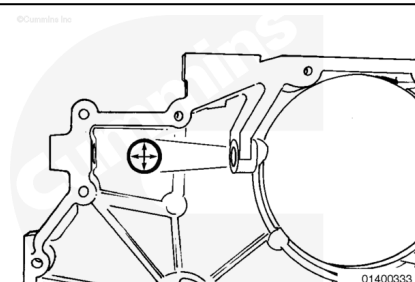


Measure

Barring Mechanism Bushing Bore.

Measure the inside diameter of the bushing bore.

Barring Mechanism Bushing Bore		
mm		in
18.215	MIN	0.717
18.265	MAX	0.719



The bushing **must** be replaced if **not** within specifications.

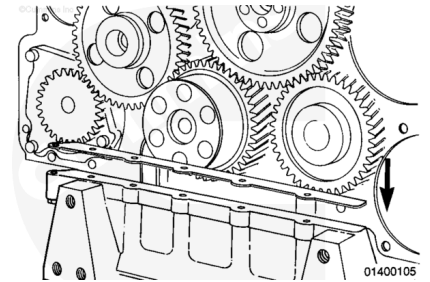
Install

Remove the old pan adapter gasket at perforations. The gasket is designed to be removed at the perforation.

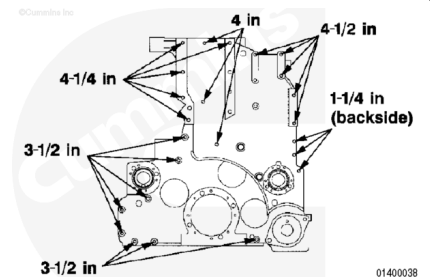
Note : The steel core gasket can not be cut without causing damage to pan adapter.

Trim the gasket to the correct width since the spacer plate is already installed.

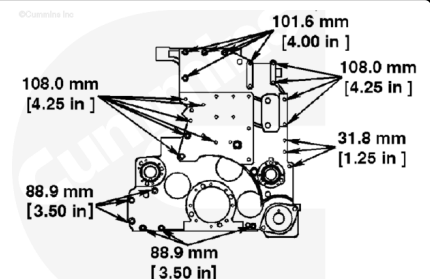
Install the service gasket, Part Number 3042591, on top of the oil pan adapter. Use a spray adhesive to hold it in position. Use Cummins® sealant, Part Number 3823494, at the joints.



This illustration shows the capscrew length requirements for the front cover on all engines, except rail applications.

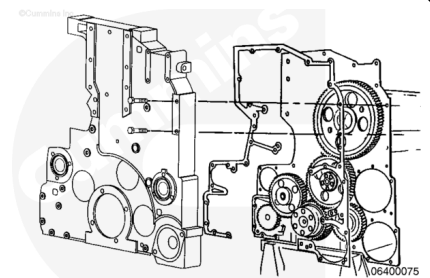


This illustration shows the capscrew length requirements for the front cover on rail application engines.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

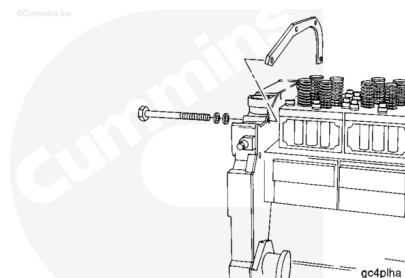


Note : Do not tighten the capscrews until the master dowel pin is installed.

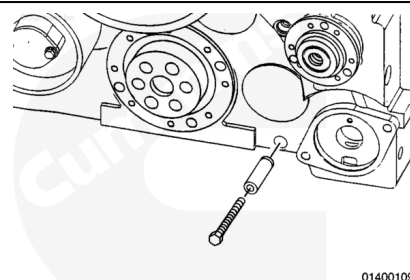
The use of guide bolts will aid the assembly.

Install the front cover gasket, cover, and capscrews.

Put the clamping plate in position and begin the installation of the capscrew.



Use a [5/16-8 x 3 inch] capscrew and a mallet. Install the threaded master dowel. Drive the dowel in until it touches the bottom of the hole in the block.



Tighten the mounting capscrews in the sequence shown.

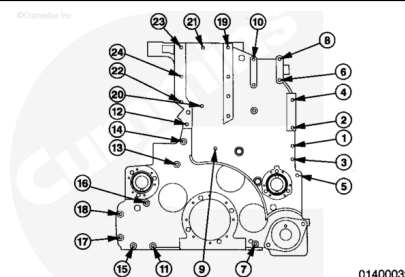
Sequence Numbers 1, 2, and 3 capscrews are located at the back of the spacer plate.

Torque Value:

Grade 5 Capscrews 45 n•m [33 ft-lb]

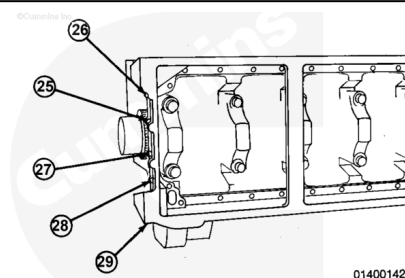
Torque Value:

Grade 8 Capscrews 61 n•m [45 ft-lb]



Install the five capscrews that attach to the bottom of the oil pan adapter. Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



Install the hydraulic pump drive.

Use clean engine oil. Lubricate the bushing in the cover.

Install the gasket, drive or cover, washers, and capscrews.

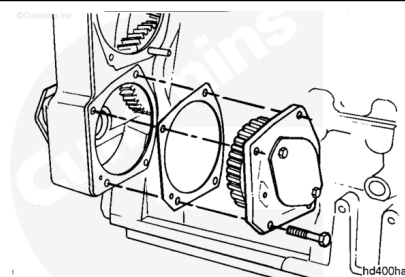
Tighten the capscrews.

Torque Value:

Grade 5 Capscrews 45 n•m [33 ft-lb]

Torque Value:

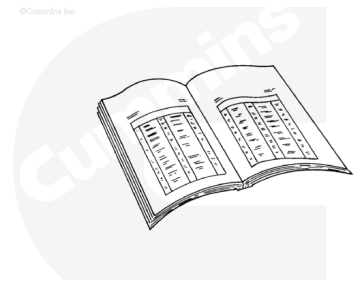
Grade 8 Capscrews 61 n•m [45 ft-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



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Note : Be sure to install the correct accessory drive assembly if an air compressor is installed.

- Install the accessory drive assembly. Refer to Procedure 009-001 in Section 9.
(/qs3/pubsys2/xml/en/procedures/20/20-009-001-tr.html)
- Install the oil scavenge pump. Refer to Procedure 007-060 in Section (/qs3/pubsys2/xml/en/procedures/20/20-007-060.html) 7 (rail applications).
- Install the front mount. Refer to Procedure 016-002 in Section (/qs3/pubsys2/xml/en/procedures/20/20-016-002-tr.html) 16 (rail applications).
- Install the fuel pump. Refer to Procedure 005-016 in Section 5. (/qs3/pubsys2/xml/en/procedures/20/20-005-016-tr.html)
- Install the front crankshaft seal. Refer to Procedure 001-023 in Section 1.
(/qs3/pubsys2/xml/en/procedures/20/20-001-023-tr.html)
- Install the vibration damper. Refer to Procedure 001-052 in Section 1. (/qs3/pubsys2/xml/en/procedures/20/20-001-052-tr.html)
- Install the thermostat housing and support. Refer to Procedure 008-015 in Section 8.
(/qs3/pubsys2/xml/en/procedures/20/20-008-015-tr.html)
- Install the accessory drive seal. Refer to Procedure 001-003 in Section 1.
(/qs3/pubsys2/xml/en/procedures/20/20-001-003-tr.html)
- Install the accessory drive pulley. Refer to Procedure 009-004 in Section 9.

(/qs3/pubsys2/xml/en/procedures/20/20-009-004.html)

- Install the fan hub idler assembly. Refer to Procedure 008-029 in Section 8.

(/qs3/pubsys2/xml/en/procedures/20/20-008-029-tr.html)

- Install the fan hub. Refer to Procedure 008-036 in Section 8. (/qs3/pubsys2/xml/en/procedures/20/20-008-036-tr.html)
- Install the alternator. Refer to Procedure 013-001 in Section (/qs3/pubsys2/xml/en/procedures/20/20-013-001-tr.html)
- Install the belt. Refer to Procedure 013-005 in Section (/qs3/pubsys2/xml/en/procedures/20/20-013-005-tr.html)
- Install the fan belt. Refer to Procedure 008-002 in Section (/qs3/pubsys2/xml/en/procedures/20/20-008-002-tr.html)
- Install the belt guard, if equipped. Refer to Procedure 008-001 in Section (/qs3/pubsys2/xml/en/procedures/20/20-008-001.html)
- Install new full-flow oil filters. Refer to Procedure 007-013 in Section 7. (/qs3/pubsys2/xml/en/procedures/20/20-007-013-tr.html)
- Connect the batteries. Refer to Procedure 013-009 in Section 13. (/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html)

Operate the engine to normal operating temperature. Check for leaks.